APOORVA AJAY

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PROFESSIONAL SUMMARY

Computational Scientist and Software Engineer with 8+ years of experience spanning scientific computing, machine learning, systems software, and large-scale computational research. Developed computational methods and automated pipelines for analyzing whole-brain microscopy datasets, supporting neuroscience research across multiple studies and contributing to peer-reviewed publications. More recently applied deep learning, GPU computing, and optimization methods to 3D cell detection, tracking, and lineage reconstruction. Experienced in Python, PyTorch, CUDA, Linux HPC, experimental validation, and translating complex research problems into reproducible computational workflows. Interested in applying this combination of empirical research, machine learning, and systems engineering to challenging problems in scientific computing, computational research, and advanced AI systems.

RESEARCH & TECHNICAL SKILLS

Machine Learning: PyTorch, 3D U-Net, Vision Transformers, Random Forest, Model Ensembling, ONNX, Test-Time Augmentation, Mixed-Precision Inference

Scientific Computing & Image Analysis: Python, NumPy, scikit-learn, MATLAB, SimpleElastix/ SimpleITK, Ilastik, Image Registration, Segmentation, 3D Reconstruction, Quantitative Image Analysis

GPU & Large-Scale Computing: CUDA, NVIDIA GPUs, Slurm, Nextflow, Linux HPC, Distributed Computing, GPU-Accelerated Workflows

Systems & Programming: Python, C/C++, Java, Perl, Bash/Shell, Linux/Unix, AIX, Systems Programming, Performance Analysis, Debugging

Software Engineering: Git, Automated Testing, Unit Testing, Code Review, CI/CD, Jenkins, Docker

Visualization: Neuroglancer, ClearVolume, Imaris, Amira

RESEARCH & PROFESSIONAL EXPERIENCE

Computational Scientist, UT Southwestern Medical Center (WBMF) **Feb 2017 – Jun 2022**

- Developed automated computational workflows for transforming large-scale Serial Two-Photon Tomography datasets into quantitative measurements for neuroscience research across multiple laboratories.
- Developed Python- and MATLAB-based methods for 3D reconstruction of two-dimensional image mosaics, preprocessing, and volumetric registration of whole-brain datasets to the Allen Common Coordinate Framework (CCFv3).
- Scaled computational workflows across Linux HPC infrastructure using Nextflow and Slurm, to support processing of large imaging datasets and research cohorts.
- Developed supervised machine-learning workflows using Ilastik to identify experiment-specific biological signals and generate probability maps for whole-dataset quantitative analysis.
- Performed dimensionality reduction and feature analysis using PCA and scikit-learn and used Neuroglancer and 3D visualization tools for interactive exploration and validation of volumetric results.
- Worked closely with experimental researchers to translate scientific questions into computational analyses, investigate failed or unexpected results, validate outputs, and document reproducible workflows.
- Trained researchers to annotate representative images, develop supervised segmentation models, execute computational workflows, and interpret resulting quantitative outputs.

- Contributed computational analysis to research spanning brain-wide connectivity, neurological disease progression, atlas generation, stroke recovery, and sleep/circadian studies, including work resulting in peer-reviewed publications.

Software Engineer, IBM India Ltd. **Oct 2011 – Jul 2014**

- Designed and developed C-based data-collection software for the IBM POWER performance-management tool PMAIX, collecting system and resource-utilization data from AIX/Linux environments.
- Developed kernel-extension-based components to query the POWER hypervisor and collect processor, memory, shared-processor-pool, and cross-LPAR performance data.
- Debugged software issues and validated system-level data collection across AIX/Linux environments and IBM POWER systems.
- Contributed across the software development lifecycle, including technical design, implementation, unit testing, code review, integration, automated regression testing, and release validation.

Cloud Consulting Intern, Ericsson Inc. **Jun 2015 – Dec 2015**

- Deployed application and Docker container workloads on a Linux-based hybrid-cloud platform.
- Integrated Git, Jenkins, Jira, and Redmine to automate development, build, and deployment workflows.
- Validated workload isolation and access- and quota-based policy enforcement, including application integration with a MySQL database hosted on AWS.

Student Worker, ARE Lab - University of Texas at Dallas **Sep 2014 – Mar 2015**

- Supported an NSF-sponsored research project investigating cloud-computing system architectures and workload performance.
- Deployed an AppScale cluster using Vagrant and Oracle VirtualBox and provisioned a MySQL database with an OLTP workload.
- Executed TPC-C Java workloads and used Python-based tooling to generate concurrent transactions and collect system-usage statistics for performance analysis.

SELECTED MACHINE LEARNING PROJECTS

Biohub – Cell Tracking During Development | Kaggle Competition **Aug 2026 – Sep 2026**

Bronze Medal · Top 8.5%

- Extended an open-source pipeline for 3D cell detection, temporal tracking, and lineage reconstruction using ensembles of pre-trained 3D U-Net and transformer models in PyTorch.
- Evaluated test-time augmentation and bidirectional edge fusion and tuned an integer-linear-programming lineage solver to improve temporal association and lineage reconstruction.
- Developed post-processing approaches to recover missed detections and reconnect broken tracks, using validation experiments and paired leaderboard submissions to evaluate changes.
- Optimized offline inference across two NVIDIA T4 GPUs using CUDA-enabled PyTorch and mixed-precision inference while operating within strict compute and runtime constraints.

Neurogolf 2026 — ARC-AGI ONNX Challenge

- Developed constrained ONNX neural-network computation graphs to solve abstract reasoning transformations under static-shape, restricted-operator, and model-size constraints.
- Translated inferred transformation rules into tensor operations and iteratively validated implementations against ARC-AGI examples and larger generated test sets.
- Produced solutions for multiple tasks that passed all available ARC-AGI examples and more than 260 generated examples per task.

EDUCATION

Master of Science in Computer Science (Data Science) Sept 2014 – Aug 2016
University of Texas at Dallas, TX
Relevant Coursework: Machine Learning, Design and Analysis of Algorithms, Advanced Operating Systems, Computer Vision, Big Data Management and Analytics, Statistical Methods for Data Science

Bachelor of Engineering in Electronics and Communications Aug 2007 – May 2011
Visvesvaraya Technological University, India

SELECTED PUBLICATIONS

- Ramirez D.M.O., Whitesell J.D., Bhagwat N., Thomas T.L., Ajay A.D., et al. (2025). Spontaneous pathology in PS19 tauopathy mice progresses via brain networks. Neurobiology of Disease. DOI
- Poinsatte K., Kenwood M., Betz D., Nawaby A., Ajay A.D., et al. (2025). SpinalTRAQ: A novel pipeline for volumetric cervical spinal cord analysis identifies the corticospinal tract synaptic projectome in healthy and post-stroke mice. eNeuro. DOI
- Poinsatte K., Betz D., Torres V.O., Ajay A.D., et al. (2019). Visualization and quantification of post-stroke neural connectivity and neuroinflammation using serial two-photon tomography. Frontiers in Neuroscience. DOI
- Sterling B.O., Torres V.O., Ajay A.D., et al. (2020). B cells migrate into remote brain areas and support recovery after focal stroke. Proceedings of the National Academy of Sciences. DOI
- Ramirez D.M.O., Ajay A.D., Goldberg M.P., & Meeks J.P.,(2019). Serial Multi-photon Tomography and Analysis of Volumetric Images of the Mouse Brain. Multi-photon Microscopy, Neuromethods series. Springer Nature

SELECTED PRESENTATIONS

- Computational Methods for Registration and Analysis of Whole Brain Volumetric Image Data Acquired via STPT. Society for Neuroscience Annual Conference, Chicago, IL, Oct 2019.
- An Automated Image Analysis Pipeline for Registration and Quantification of Volumetric STPT Images in Animal Models of Brain Injury and Disease. Society for Neuroscience Annual Meeting, San Diego, CA, Nov 2018.